DEM B2B SUMMIT SCHEDULING ENGINE
PHASES 1 TO 3 TECHNICAL IMPLEMENTATION PLAN

Lead Architect: Sherif Seif Eldin
UK/EU Timezone Overlap | London Staging Ready
Live Production Engine: autobathmelbourne.com

01. UNDERSTANDING OF PHASES 1 TO 3

- **Phase 1 (Core Admin Engine):** Replaces manual spreadsheet and Typeform friction with an automated scheduler. Organizers configure event parameters (days, hours, meeting slot lengths, keynote blackout times), upload participant quotas, ingest Typeform preference rankings, generate conflict-free timetables, execute manual drag overrides with rule blockers, and manage live-event no-shows.
- **Phase 2 (Personal Schedule View):** View-only portal for sponsors & buyers to view finalized schedules, download branded PDFs, and receive instant reschedule alerts via Resend email.
- **Phase 3 (Single Login Portal):** Unified authentication environment where attendees submit meeting rankings and access finalized agendas in one place.

02. TECH STACK & UK/GDPR COMPLIANCE

- **Frontend:** Next.js 15 (App Router), Tailwind CSS, Lucide Icons, and React-PDF for client-side timetable downloads. Fast, responsive, and mobile-ready.
- **Database & Backend:** Next.js Server Actions with Zod validation + Supabase PostgreSQL hosted in **AWS London (eu-west-2)** for strict UK Data Protection & GDPR compliance.
- **Webhooks & Comms:** Typeform Webhook API (auto-syncing participant preferences) + Resend API (sub-second transactional reschedule alerts).

03. SCHEDULING ALGORITHM & RULE ENFORCEMENT

- **Deterministic Constraint Solver:** Engineered in TypeScript/Node.js utilizing greedy & backtracking heuristics. The engine prioritizes contracted sponsor quotas (Tier 1 first), mutual ranking weights, availability windows, room capacities, and keynote blackout periods.
- **Real-Time Atomic Rule-Blocker:** Manual drag-and-drop modifications by organizers are validated against the *Meeting Scheduling Rules* in real time, automatically blocking any assignment that causes double-bookings or violates contracted minimums.
- **Live Event No-Show Handler:** Instant one-click slot release with intelligent replacement candidate surfacing during live event days.

04. AI-NATIVE WORKFLOW & CODE AUDIT

- **Agentic AI Velocity:** Utilizing Claude Code & Antigravity IDE loops for rapid schema scaffolding, API route generation, and test generation.
- **Human & Test Verification:** 100% of scheduling engine logic, quota edge cases, and tie-breaking algorithms are verified via comprehensive automated unit & stress test suites (Jest/Vitest) prior to organizer review.
- **Weekly Staging Demos:** Continuous Vercel preview builds with active stakeholder testing.

05. FIXED MILESTONES, TIMELINE & DELIVERABLES ($3,000 USD TOTAL)

Phase & Focus	Timeline	Core Deliverables & Specifications	Fixed Price
Phase 1: Scheduling Engine Core Admin System	Weeks 1–4 (Nov – Dec)	Event setup, Typeform webhook ingestion, constraint scheduling solver, admin timetable review, manual override rule-blocker, live event no-show handler, CSV/PDF export. Weekly staging demos.	**$1,500**
Phase 2: Personal Schedule Attendee Access	Weeks 5–6 (Dec – Jan)	Participant view-only portal, personalized branded PDF timetable generator, automated Resend email reschedule notifications.	**$800**
Phase 3: Single Login Portal Production Launch	Weeks 7–8 (Jan – Feb)	Unified attendee authentication (preference submission + schedule viewing), stress testing (500+ attendees), edge deployment & handover.	**$700**

ZERO DOUBLE-BOOKINGS: Deterministic atomic locks validate every slot modification in real-time.

UK/GDPR COMPLIANCE: Supabase PostgreSQL hosted directly in AWS London (eu-west-2).

PRODUCTION TRACK RECORD: Multi-slot scheduling engine live at autobathmelbourne.com.

Availability: Available for immediate November kickoff | 30+ hrs/week overlap with UK/London business hours.
Delivery Commitment: Weekly live staging demos on Vercel preview environments with full source code handover.

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